

Experiment Name:

Software Configuration Management with Git and GitHub

Objective:

- To understand the fundamental concepts and importance of Software Configuration Management (SCM) and version control.
- To learn and practice basic, essential Git commands for local version control.
- To gain experience with remote repositories by using GitHub
- To implement and understand the "branch and merge" workflow by creating feature branches

Theory:

Git is a distributed version control system widely used in modern software development. It enables developers to track code changes, restore previous versions, and collaborate efficiently with others.

GitHub is an online platform that hosts Git repositories and offers collaboration tools such as pull requests and issue tracking. In this experiment, I will explore the fundamental Git workflow by creating a local repository, making commits, and pushing your work to a remote GitHub repository.

Required Equipment / Software

- A computer with an internet connection.
- Git installed on your system.
- A free GitHub account.
- A text editor or IDE

Procedure Followed:

1. **Install Git**
2. **Configure Git:**

```
git config --global user.name "Md. Asad-Al-Adil"  
git config --global user.email "adilasadals@gmail.com.com"
```

3. **Create a Project:** A HTML Portfolio website is created along with index.html
4. **Create GitHub Repository:** A repository is created named **asadaladil.github.io**

1
General

Owner *

asadaladil

Repository name *

asadaladil.github.io

asadaladil.github.io already exists in this account

Great repository names are short and memorable. How about **ubiquitous-octo-eureka**?

Description

0 / 350 characters

2
Configuration

Choose visibility *

Choose who can see and commit to this repository

Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

5. Push to GitHub:

```
PS F:\RUET CSE LAB\3206-Software Engineering\1. 14-10-2025\asadaladil.github.io> git add .
PS F:\RUET CSE LAB\3206-Software Engineering\1. 14-10-2025\asadaladil.github.io> git commit -m "first commit"
[version1 8316c77] first commit
1 file changed, 24 insertions(+), 23 deletions(-)
PS F:\RUET CSE LAB\3206-Software Engineering\1. 14-10-2025\asadaladil.github.io> git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 545 bytes | 545.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/asadaladil/asadaladil.github.io.git
c8d53a1..8316c77 version1 -> version1
```

8. Verify: Refresh your GitHub repository page.

asadaladil first commit	8316c77 · 7 minutes ago	6 Commits
Adil.jpg	27-10-2025	3 weeks ago
index.html	first commit	7 minutes ago

Branching and Merging

1. Setup:

```
git init
git add .
git commit -m "Initial commit"
```

2. Create a Feature Branch: Create a new branch named version1.

```
git branch version1
git checkout version1
```

3. Implement the Feature: In the version1 branch modifies the index.html and commit the changes.

```
# ... after modifying the file ...
git add index.html
git commit -m "Updated Portfolio website"
```

4. Merge the Feature: Switch back to the main branch and merge the feature branch into it.

```
git checkout main
git merge version1
```

```
PS F:\RUET CSE LAB\3206-Software Engineering\1. 14-10-2025\asadaladil.github.io> git branch
main
* version1
```

Forking and Pull Request

1. **Fork a Repository:** Find a public repository on GitHub and click the "Fork" button. This creates a personal copy of the repository under your account.



2. **Clone the Fork:** Clone your forked repository to your local machine.



 asadaladil Merge branch 'main' into main 42701e4 · last month 5 Commits

names.txt

2103064

last month

3. Push to Your Fork: Push the new branch to your forked repository on GitHub.

Code Blame 3 lines (3 loc) · 120 Bytes

Raw Copy Download Edit

```
1 Roll- 2103064
2 Repository Link- https://github.com/asadaladil/asadaladil.github.io
3 website- https://asadaladil.github.io
```

4. Create a Pull Request (PR): Go to your forked repository on GitHub. You will see a prompt to create a pull request. Click it, add a title and description for your changes, and submit the PR. This asks the original repository's owner to "pull" your changes into their project.

 This branch is 4 commits ahead of MotiurRahmanSany/software_engineering_lab_01:main .
Open a pull request to contribute your changes upstream.

Open pull request

 Add a title

new pull request

Add a description

Write Preview

H B I          

Hello

 

 Markdown is supported  Paste, drop, or click to add files

☒ Allow edits by maintainers  Create pull request

new pull request #1

 asadaladil wants to merge 4 commits into MotiurRahmanSany:main from asadaladil:main

 Conversation 0  Commits 4  Checks 0  Files changed 1

 asadaladil commented now

Hello



2103064

Verified

2b5d6e6

Merge branch 'main' into main

Verified

42701e4

names.txt

Verified

e2573e0

✓

No conflicts with base branch
Changes can be cleanly merged.

Still in progress? [Convert to draft](#)

Github Server:

GitHub Pages

GitHub Pages is designed to host your personal, organization, or project pages from a GitHub repository.

Your site is live at <https://asadaladil.github.io/>
Last [deployed](#) by [asadaladil](#) 5 minutes ago

[Visit site](#)[Unpublish site](#)

Build and deployment

Source

Deploy from a branch ▾

Branch

Your GitHub Pages site is currently being built from the version1 branch. [Learn more about configuring the publishing source for your site.](#)

🔗 version1 ▾

📁 / (root) ▾

Save

----After Hosting the server----

Hi, I'm Md. Asad- Al-Adil

Computer Architecture
VLSI Technology
Artificial Intelligence
Machine Learning

A Motivated Computer Engineering student and want to be specialized in VLSI design and hardware acceleration. Seeking a research-based Master's to explore low-power circuit optimization and FPGA-based architectures. Passionate about contributing to rapidly growing cutting-edge semiconductor technologies.

[View My Work](#)[Get In Touch](#)

Conclusion:

GitHub is the backbone of modern development. It bridges the gap between code management and team collaboration, ensuring that software is built faster, safer, and more transparently.